

MetAgent case study

compound_only_enrich_mammalian_RAMP_P_000000106_seed2

Input metabolite set -> pathway evidence -> final metabolic interpretation

Input

Differential metabolites (n=9)

Dihydrobiopterin

C9H13N5O3

C02953

5,6-Dihydroxyindole

C8H7NO2

C05578

3,4-Dihydroxyphenylacetaldehyde

C8H8O3

C04043

Copper

Cu

C00070

Pyrocatechol

C6H6O2

C15571

Other differential metabolites

Eicosapentaenoic acid (C06428), Sucrose (C00089), 1-Pyrroline-5-carboxylic acid (C03912), Aspirin (C01405)

Top pathway

Tyrosine metabolism

KEGG map00350 | FDR=6.61e-11
fold-enrichment=435.1 | matched 5/9

Final verifier totals

supported 18 | unsupported 0 | contradicted 1
unverifiable_v0 12

Final MetAgent report

Verifier-colored complete narrative

supported

unsupported

contradicted

unverifiable_v0

Revised Narrative

The enrichment analysis strongly supports Tyrosine metabolism (KEGG hsa00350) as the top pathway (FDR = 0.0, fold-enrichment ~ 435).

Five compounds map to this route - dihydrobiopterin (C02953), 5,6-dihydroxyindole (C05578), 3,4-dihydroxyphenylacetaldehyde (C04043), pyrocatechol (C15571), and copper (C00070) - reflecting perturbations at multiple nodes spanning biosynthetic, catabolic, and cofactor-related branches of the network.

The key drivers are dihydrobiopterin (C02953), 5,6-dihydroxyindole (C05578), and 3,4-dihydroxyphenylacetaldehyde (C04043).

Dihydrobiopterin is the fully reduced form of tetrahydrobiopterin (BH4), an essential cofactor within the tyrosine metabolism network [retracted: not supported by RaMP/KEGG].

Its differential abundance suggests altered BH4 turnover or regeneration.

3,4-Dihydroxyphenylacetaldehyde occupies the immediate downstream product of dopamine oxidation by monoamine oxidase; its elevation implies increased dopamine turnover or reduced aldehyde dehydrogenase flux.

5,6-Dihydroxyindole, an intermediate in melanin biosynthesis, also reflects shunting of dihydroxyphenylalanine-derived substrates away from neurotransmitter synthesis toward pigment pathways.

Biologically, these changes are consistent with oxidative stress or altered monoamine oxidase activity in the treatment condition.

The presence of pyrocatechol (a direct o-hydroxylation product of phenol metabolism) and copper (a cofactor for dopamine beta-hydroxylase and ceruloplasmin) reinforces oxidative and metal-dependent processing of tyrosine-derived metabolites.

The differential abundance of multiple downstream intermediates (dihydrobiopterin, 3,4-dihydroxyphenylacetaldehyde, 5,6-dihydroxyindole) within Tyrosine metabolism suggests increased flux through this pathway in the treatment group.

In summary, the data indicate perturbation of Tyrosine metabolism through both the neurotransmitter and melanin branches, driven by elevated dihydrobiopterin, 3,4-dihydroxyphenylacetaldehyde, and 5,6-dihydroxyindole, with plausible downstream consequences for catecholamine homeostasis and oxidative balance [retracted: enhanced catabolism - not a valid pathway name].

Limitations: RaMP-DB enrichment analysis identified Tyrosine metabolism (KEGG hsa00350) as the top pathway, but the input list contained only downstream metabolites.

Neither tyrosine nor L-DOPA was directly measured, so the upstream driver (substrate availability vs. enzymatic regulation) cannot be determined from the data alone.

Additionally, enzyme-level claims (e.g., tyrosine hydroxylase rate-limiting step) lack direct pathway-database support and were retracted accordingly.